Outage Duration, Capital, and Firm Backup-Power Choice in Developing Countries

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Abstract

Unreliable electricity forces firms in developing economies to secure their own power, a need historically met by diesel generators. On-site solar is now broadly cost-competitive with diesel self-generation over its lifetime, yet the generator still dominates. I develop a model of the firm's backup-power choice in which the two technologies differ in two respects. A generator is dispatchable and can be matched to an outage, and the model implies a threshold outage length beyond which running it pays, which makes outage duration rather than frequency the operative margin. Solar without storage is intermittent, and its large upfront cost makes adoption depend on firm capability and access to finance rather than on outage exposure. I test these predictions with firm-level data from the 2025 and 2026 World Bank Enterprise Surveys for 11,252 firms in 36 developing economies, with 17 high-income economies as a benchmark. I use linear probability and multinomial logit models with country and sector fixed effects, country-clustered standard errors, and the wild-cluster bootstrap. Omitted local conditions,

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measurement error in self-reported outages, and reverse causality could each bias these estimates. As a robustness check, I instrument outage exposure with the outages reported by other firms in the same locality, with and without local-development controls. The findings show that generator ownership rises with outage duration, while solar adoption responds weakly to outages and instead tracks firm size, exporting, and digital presence. The paper makes three contributions. It extends the literature on self-generation, which has treated backup power as fossil generation, to the choice between diesel and solar. It identifies outage duration, not frequency, as the margin that drives generator investment. It separates the dispatchability and capital channels, which explains why falling panel prices have not displaced the generator. Policies that lower the upfront cost of solar would raise clean self-generation but would not by themselves move firms off generators unless they also finance storage.

Keywords: power outages, self-generation, solar adoption, credit constraints, dispatchability, developing economies

JEL: Q42, D22, O13, L94, Q56

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1. Introduction

Unreliable electricity forces firms in developing economies to secure their own power, a need historically met by diesel generators (Steinbuks and Foster, 2010). The economics of this response, however, have changed markedly. The installed cost of on-site solar has fallen substantially, and on a levelised-cost basis solar is now broadly competitive with both new fossil-fuel generation and diesel self-generation (Lam et al., 2019; IRENA, 2024). A comparison of lifetime costs would therefore suggest that firms exposed to outages should increasingly substitute away from diesel generators toward the cleaner

and potentially lower-cost solar technology. Yet the data point in the opposite direction.

Across 36 developing economies in the most recent World Bank Enterprise Surveys, 40.0 percent of firms own a generator, compared with only 10.9 percent that use on-site solar. Among firms with either technology, generators supply 10.7 percent of electricity, versus 3.9 percent from solar. The gap is not explained simply by firms' exposure to outages. Among firms reporting no outages, 30.1 percent hold some form of backup capacity, compared with 61.4 percent of firms experiencing long outages (Figure 1). Yet over the same range, generator ownership rises from 24.9 to 53.5 percent, while solar adoption increases only from 10.3 to 15.6 percent. The cross-country pattern shows the same. Solar adoption has no apparent relationship to grid reliability (Figure 2). In the Netherlands, Austria, and Germany, where relatively few firms report outages, between 27.6 and 35.5 percent of firms use on-site solar. By contrast, only 0.6 to 1.4 percent of firms in Ethiopia, Guinea, and Mozambique adopt solar, where outages are widespread.

The stakes are substantial. Outages depress output and force firms into costly self-supply, with backup generation costing roughly three times the price of grid electricity (Allcott et al., 2016; Steinbuks and Foster, 2010), and diesel generation adds local air pollution and carbon emissions (Lam et al., 2019). The pattern becomes less puzzling once generators and solar panels are understood as responses to different constraints rather than as direct substitutes.

The paper investigates firms' backup-power choice using firm-level data from the most recent rounds of the World Bank Enterprise Surveys (WBES). For each establishment, the survey records whether it generates electricity from solar panels and the share of electricity those panels supply. The survey

also reports generator ownership in addition to the number and duration of reported grid outages. The main estimation sample pools 11,252 firms across 36 developing economies in six regions. A separate benchmark of approximately 9,360 firms across 17 high-income economies is used only to examine the determinants of solar adoption. The analysis is organised around two mechanisms that the model in Section 3 treats as distinct.

The first mechanism is dispatchability. A diesel generator can be switched on when the grid fails, whereas solar without storage produces according to the availability of sunlight. A firm's exposure to outages should therefore be more strongly associated with adoption of dispatchable generation than with adoption of solar. The second mechanism is capital intensity. Solar is a capital-intensive investment. Lifetime costs are paid upfront, with relatively low operating costs. A diesel generator, by contrast, requires a smaller initial investment but entails substantial recurring fuel costs and maintenance. Solar adoption as a self-generation investment should therefore depend more on firms' capabilities and access to finance than on their exposure to outages. The paper tests three hypotheses that follow from these mechanisms. First, firms exposed to outages should be more likely to own a diesel generator than to adopt solar. Second, the probability of generator ownership should increase more strongly with outage duration than with outage frequency. Third, solar adoption should be more strongly associated with firm capability and access to capital than with exposure to outages.

The central result is an asymmetry in firms' responses to unreliable electricity. Outage exposure is strongly associated with generator ownership, more with outage duration than frequency. Solar adoption instead tracks firm capability and access to capital. The estimates are conditional associations from a single cross-section. Omitted local conditions, measurement

error in self-reported outages, and reverse causality could each bias them. Section 4.6 discusses each source and describes an instrumental-variable robustness check.

This paper makes three contributions. First, it extends the literature on self-generation to the choice between the diesel generator and solar. That literature documents that firms respond to unreliable power by installing private generators (Reinikka and Svensson, 2002; Steinbuks and Foster, 2010; Oseni and Pollitt, 2015) but treats self-generation as fossil generation. This paper brings multi-country, firm-level evidence to the choice between the two technologies and identifies the mechanisms from variation within country. Second, it identifies outage duration rather than frequency as the operative margin for generator investment, a distinction lost when exposure is summarised in a single index. Third, it separates the dispatchability and capital channels. This links the evidence that upfront cost holds back solar take-up (Lee et al., 2020; Burgess et al., 2023) to firms' backup choices and clarifies why falling solar panel prices have not shifted the firm's backup-power choice. Section 2 reviews the literature on unreliable power and self-generation, Section 3 sets out a model of the firm's backup-power choice, Section 4 describes the World Bank Enterprise Survey data and the empirical strategy, Section 5 presents the results, and Section 6 concludes with the policy implications.

2. Related Literature

2.1. Electricity Unreliability and Firm Performance

A large body of evidence establishes unreliable electricity as a first-order constraint on firm performance in developing economies, working through both the price of supply and its reliability. Survey-based accounting places

indirect costs related to infrastructure and services below 15 percent of total firm costs in stronger-performing developing economies. In most of Sub-Saharan Africa, these costs reach 20 to 30 percent, with energy the single largest component (Eifert et al., 2008). Where the constraint has been studied with credible identification, the output consequences are sizeable. Drawing on plausibly exogenous variation in electricity shortages across Indian states, Allcott et al. (2016) show that shortages reduce plant revenues and producer surplus with only modest effects on measured productivity as plants substitute toward stored inputs. Evidence from Chinese industry reveals a different margin of the same constraint. Fisher-Vanden et al. (2015) find that firms facing greater electricity scarcity re-optimise toward purchased inputs. This raises production costs by about 8 percent while limiting the measured productivity losses they would otherwise bear. The pattern recurs across Africa, where Cole et al. (2018) link outages to lower firm productivity and sales. Mensah (2024) traces the employment consequences of chronic shortages, while firm-level analyses attribute a substantial part of the region's total factor productivity gap to weak infrastructure, including power (Arnold et al., 2008). Related work shows that the price and availability of power shift firms toward less electricity-intensive and lower-growth activities (Abeberese, 2017). Among micro-firms in Ghana, those without employees lose revenue and profits during outages, while firms with employees adjust hours and wages rather than output (Hardy and McCasland, 2021). At the macroeconomic level, the same unreliability lowers growth (Andersen and Dalgaard, 2013). Beyond these direct effects, electricity constraints also shape firms' long-run investment decisions (Alby et al., 2013). The consistent finding across these settings is that outages impose substantial and unevenly distributed costs, which gives firms a strong private incentive to

insure themselves against unreliable supply.

2.2. Self-Generation and the Economics of Backup Power

The standard firm response to an unreliable grid is to self-generate, a response documented throughout the literature. Reinikka and Svensson (2002) show that poor public power induces firms to substitute into private generators, crowding out more productive investment. This finding frames self-generation as costly insurance rather than as a productive choice. Foster and Steinbuks (2009) document that own generation accounts for around 6 percent of the overall installed capacity in Sub-Saharan Africa, roughly 12 percent in low-income and post-conflict economies, and more than 20 percent in countries where the grid is weakest. Steinbuks and Foster (2010) provide the most direct precedent for the present paper, modelling in-house generation across 25 African economies. They find that generator ownership rises with outage exposure, firm size, age, and export orientation. Their evidence suggests that ownership would remain near 20 percent even under a perfectly reliable grid, and that self-generated power costs US$0.30 to 0.70 per kilowatt-hour, which is about three times the price of grid electricity. Oseni and Pollitt (2015) extend this accounting to the outage losses that remain after a firm has installed backup. They show across a large sample of African firms that firms without backup could have cut their outage losses by roughly 6 to 46 percent by self-generating. However, firms that self-generate still bear sizeable unmitigated losses, as a generator rarely covers a firm's entire load. This literature treats self-generation as synonymous with fossil generation, a view established before the decline in solar costs. If solar can substitute for the diesel generator in the backup role, given that it is cheaper on a levelised-cost basis, outage exposure should drive solar adoption as it

drives generator ownership.

2.3. Distributed Solar Adoption and the Capital Barrier

Distributed solar has continued to fall in cost, and on a levelised-cost basis it is broadly competitive with diesel self-generation in high-outage settings (IRENA, 2024). Two features of the technology nonetheless complicate its use as backup. Solar output is intermittent and follows the sunlight rather than the timing of an outage, and without storage it cannot be dispatched to the moment the grid fails. The systems literature treats dispatchable and variable sources as substitutes that compete on a cost basis rather than as complements, such that a fall in the cost of one displaces the other (Yuan et al., 2020). Solar also carries a large upfront cost, and field evidence on off-grid solar and rural electrification shows that take-up is held back by the ability and willingness to meet that cost, with demand falling sharply as the price rises (Lee et al., 2020; Burgess et al., 2023). Socioeconomic gains are also limited where only basic access is provided (Aklin et al., 2017). Two strands of the firm literature speak to this friction. Access to finance is among the most binding constraints on investment in low-income economies (Beck et al., 2005). The capability to absorb and operate advanced technology is concentrated among larger, more productive, and more internationally connected firms (Verhoogen, 2023). These two features map onto the two mechanisms the paper separates. Solar without storage cannot be dispatched, and its adoption should therefore not respond to outage exposure. Solar is capital-intensive, and its adoption should therefore track capability and finance. No existing study separates these two channels with firm-level data spanning many developing economies.

3. A Model of the Backup-Power Choice

This section develops a model of how a firm chooses backup power when the grid is unreliable. The model treats self-generation as an investment to mitigate outage losses (Steinbuks and Foster, 2010; Oseni and Pollitt, 2015). Solar in this model differs from the diesel generator in two respects. Without storage, it cannot be dispatched to an outage, and it carries a high upfront cost.

Consider a firm that draws electricity from an unreliable grid. Outages are described by two parameters, the number of outages in a period, n, and the typical duration of an outage in hours, h. Each hour without power costs the firm lost output, an amount that rises with the electricity dependence of its operations. Let this per-hour loss be denoted by the term in Equation 1, where Q is the value of output the firm would produce in an hour of normal operation and the parameter λ, bounded between zero and one, is the share of that value lost when power is interrupted, a measure of the firm's vulnerability to outages.

$$\ell_i = \lambda_i Q_i \tag{1}$$

With no backup, the firm's expected loss over the period is the per-hour loss multiplied by the total hours without power, given in Equation 2. Total outage hours are the product of frequency and duration, and in the absence of backup the two enter the loss symmetrically.

$$\Lambda^0 = \ell_i n_i h_i \tag{2}$$

The firm can mitigate this loss by installing one or both of two backup technologies. The diesel generator carries a capital cost that is low relative to its capacity, denoted F_D, and a fuel cost per hour of operation, denoted

c_D. The generator is dispatchable, producing exactly when the firm switches it on, and can therefore be matched to the outage. Deploying the generator within a given outage carries a small fixed cost, γ, for switching over, fuelling, and maintaining readiness. The generator is worth running in a given event only when the loss it avoids, net of fuel, exceeds this fixed cost, which occurs only once the outage is sufficiently long. The annualised net value of owning a generator is given in Equation 3, where r annualises the capital cost.

$$V_i^D = n_i \max\{(\ell_i - c_D)h_i - \gamma,\, 0\} - rF_D \tag{3}$$

The specification assumes that the fuel cost exceeds the grid price, which rules out running a generator to displace grid power and holds empirically given that self-generation costs several times the grid price. The generator is therefore worth owning only for firms whose per-hour loss exceeds the fuel cost.

Within a given outage, the firm operates the generator only once the avoided loss, net of fuel, exceeds the fixed deployment cost γ. Equating the two yields the threshold duration in Equation 4. A longer typical outage raises the generator's value directly, whereas outages shorter than the threshold contribute nothing, and the cost γ is incurred once per event. The value of ownership is therefore more elastic with respect to typical duration than to frequency, the model's second prediction, although both raise the generator's value.

$$\begin{aligned} \ell_i h_i - c_D h_i - \gamma &= (\ell_i - c_D)h_i - \gamma \\ (\ell_i - c_D)h_i - \gamma &> 0 \\ (\ell_i - c_D)h_i^* &= \gamma \\ h_i^* &= \frac{\gamma}{\ell_i - c_D} \end{aligned} \tag{4}$$

Solar generation reverses the cost structure and adds a timing problem. Its capital cost is high, denoted F_S and well above the diesel generator's, while its marginal cost is near zero. Its output, however, is intermittent, following the sunlight rather than the timing of outages, and without storage it cannot be dispatched to the moment of need. The value of solar therefore has two parts, given in Equation 5. The first is the saving from displacing grid electricity at price p over annual solar output G, scaled by a utilisation term θ that rises with the firm's scale and capability. The second is the outage coverage that solar provides incidentally, scaled by s, the probability that solar is generating at the moment the grid fails. Without storage this probability is bounded well below one and lies outside the firm's control, as solar output cannot be timed to a given event.

$$V_i^S = pG_i\theta_i + s_i\Lambda^0 - rF_S \tag{5}$$

Storage changes the timing problem without changing the cost structure. A battery paired with the panels stores daytime output and releases it at the moment the grid fails, raising the coverage probability to one and making clean self-generation dispatchable in the same sense as the generator. The value of a solar-plus-storage system is given in Equation 6, where the coverage term now enters in full and the marginal cost remains near zero. The capital cost, denoted F_{S+B}, is the sum of the panel cost and the cost of storage capacity.

$$V_i^{S+B} = pG_i\theta_i + \Lambda^0 - rF_{S+B}, \qquad F_{S+B} = F_S + F_B \tag{6}$$

The hybrid system's capital cost exceeds the standalone-solar cost, which in turn stands well above the generator's, and the adoption rule of Equation 7 applies unchanged. Two implications follow. With storage, the coverage

term enters in full and the value of clean backup rises with outage exposure as the generator's does, which shows that the predicted asymmetry reflects dispatchability and capital rather than clean generation as such. The hybrid system also faces the largest capital hurdle, which binds most tightly in the capital-scarce settings where outages are most severe, and a firm that faces long outages but cannot finance it turns to the low-cost generator. Neither panel-only nor storage-backed systems should therefore respond to outage exposure in the data.

Given that s is bounded below one and cannot be controlled by the firm, outage exposure raises the value of solar only weakly, whereas it raises the value of the generator directly, a result independent of the cost of solar. The value of solar depends instead on the displacement saving, which grows with firm scale and capability, and on the firm's ability to finance the upfront cost. This is the capital channel, summarised in Equation 7, where A denotes the firm's internal funds and credit access. The financing constraint binds for solar among firms with limited capital but rarely for the low-cost generator. Where capital is broadly scarce, it lowers the overall level of solar adoption rather than separating more and less constrained firms within an economy, and the channel is identified from the contrast between capital-scarce and capital-abundant settings.

$$V_i^j \geq 0$$

$$F_j \leq A_i$$

$$\text{adopt } j \iff V_i^j \geq 0 \quad \text{and} \quad F_j \leq A_i \tag{7}$$

The model yields three predictions, and each is tested by a different part of the analysis. First, the probability of owning a generator rises with outage exposure, while the probability of adopting solar rises only weakly with

exposure. This asymmetry follows from dispatchability and the near-zero outage-coverage probability in Equation 5. Second, the generator response is stronger for outage duration than for frequency, a pattern driven by the threshold in Equation 4, which makes long outages the events worth covering. Third, solar adoption increases with firm capability, which raises the utilisation term, and with access to capital, which relaxes the financing constraint, and this holds whether or not the firm faces outages.

4. Data and Empirical Strategy

4.1. The World Bank Enterprise Surveys and the Solar Module

The analysis uses data from the World Bank Enterprise Surveys (WBES). WBES conducts nationally representative firm-level surveys administered with a standardised questionnaire that covers formal private firms in manufacturing and services. The 2025 and 2026 rounds carry an on-site solar module that asks whether the establishment used on-site solar panels over the fiscal year, with a follow-up recording the share of the firm's electricity that solar supplied. The module sits alongside the long-standing items on outage frequency, outage duration, and generator ownership. The module captures solar as an on-site self-generation technology, though it does not ask whether solar was installed as outage backup, and the survey records no battery storage. The generator items, by contrast, record a device whose purpose is backup. The analysis therefore compares a technology installed for backup with one installed for self-generation, and the asymmetry documented below partly reflects the distinct functions of the two technologies. The analysis covers 36 developing economies where the module is available at the time of the analysis—18 in Sub-Saharan Africa, 6 in Latin America

and the Caribbean, 5 in East Asia and the Pacific, 3 in South Asia, 2 in the Middle East and North Africa, and 2 in Europe and Central Asia. In addition, 17 high-income economies are used as a benchmark, where solar adoption is recorded. The solar question was unavailable for two countries, India and Poland, which are therefore excluded from the sample. The final estimation sample comprises 11,252 firms, with the high-income economies entering only the analysis of the determinants of solar adoption. Table A.1 reports the per-country pattern of adoption and exposure.

4.2. Backup Adoption and Energy Outcomes

The two main adoption outcomes are binary. Generator ownership equals one if the establishment owned or shared a generator over the last fiscal year (WBES question c10), and zero otherwise. The survey does not record fuel type, although most such units are diesel (Lam et al., 2019). Solar adoption equals one if the establishment used on-site solar panels to generate electricity over the last fiscal year (c43), and zero otherwise. Two continuous outcomes measure the percentage of the establishment's electricity supplied by its generator (c11) and by its solar panels (c44), on a scale of 0 to 100, with non-adopters assigned zero. A four-category backup portfolio combines the two indicators and classifies each firm as holding neither technology, a generator only, solar only, or both, which captures the joint adoption decision directly rather than treating one technology as a determinant of the other. Finally, two indicators from the survey's green economy module test whether unreliable electricity induces a broader energy response. Energy management equals one if the firm reports adopting energy management measures (ge8d), and carbon monitoring equals one if it monitors its own carbon dioxide emissions (ge7). Appendix Table B.1 lists every variable with

its survey question and coding.

4.3. Outage Exposure

Outage exposure is measured by two variables that enter every specification together, both from the survey's infrastructure module. Firms first report whether the establishment experienced power outages in the last fiscal year (c6). Those that did report the number of outages in a typical month (c7) and how long a typical outage lasted, in hours and minutes (c8). Outage frequency is the number of outages in a typical month, and outage duration is the typical length of an outage in hours. Firms reporting no outages are assigned zero on both measures, which keeps them in the sample rather than selecting on having outages. The two measures are only modestly correlated, at about 0.23, and capture distinct features of an unreliable grid, the length of a typical interruption and how often interruptions occur. Entering them jointly is central to the paper, given that the model predicts the two margins matter differently for the diesel generator, a distinction lost when exposure is summarised by a single index.

4.4. Firm Characteristics

All specifications condition on a compact set of firm characteristics that are standard correlates of technology adoption. Firm size is the natural logarithm of the number of permanent full-time employees at the end of the last fiscal year (l1). Firm age is the natural logarithm of one plus the number of years the establishment has been operating (b5). Exporter equals one if direct exports account for more than 0 percent of the establishment's sales (d3c). Foreign ownership equals one if private foreign individuals, companies, or organisations own more than 10 percent of the firm (b2b). Capital city

equals one if the establishment is located in the country's capital city (a3b). This baseline control set follows comparable WBES studies of self-generation (Steinbuks and Foster, 2010).

A broader set of capability and finance variables enters the heterogeneity and solar-adoption analyses. Quality certification equals one if the establishment holds an internationally recognised quality certification, such as ISO 9000 (b8). Website equals one if the establishment has its own website (c22b). Manager experience is the natural logarithm of one plus the top manager's years of experience working in the firm's sector (b7). Training equals one if the establishment ran formal training programmes for its permanent full-time employees in the last fiscal year (l10).

The finance measures follow the definitions of the World Bank Enterprise Surveys Indicators Database, which constructs harmonised firm-level indicators under a standard published methodology. A firm is credit constrained if it is fully or partially credit constrained under that methodology. Fully constrained firms have no external loan and either applied for one and were rejected or did not apply for reasons other than having sufficient capital, such as unfavourable terms or complex procedures. Partially constrained firms have some external finance but were rejected or discouraged when seeking more. Unconstrained firms either did not need a loan or applied and were approved. Bank credit equals one if the establishment has a loan or line of credit from a financial institution. Among firms that purchased fixed assets in the last fiscal year (k4), external investment finance equals one if any part of that purchase was financed from sources other than internal funds. The finance obstacle is the answer to the question "To what degree is access to finance an obstacle to the current operations of this establishment?" (k30), coded 0 for no obstacle, 1 for a minor obstacle, 2 for a moderate obstacle, 3

for a major obstacle, and 4 for a very severe obstacle, and is used for the descriptive comparison in Table A.2. At the country level, credit depth equals one if domestic credit to the private sector as a share of GDP, from the World Development Indicators for the most recent available year, lies above the median of the developing-country sample. Country and sector fixed effects absorb remaining institutional and industry-level variation. Responses of "don't know" and refusals, and values outside plausible ranges, are coded as missing.

4.5. Descriptive Statistics

Table 1 reports summary statistics for the estimation sample. Diesel generators are common, with about 40 percent of firms owning one, whereas on-site solar remains rare at about 11 percent, highlighting the adoption asymmetry examined in the paper. Generators supply about 11 percent of electricity on average, compared with about 4 percent from solar. The typical firm reports outages lasting about two hours and occurring roughly five times per month, with both measures strongly right-skewed. The average establishment employs around 13 workers; about 9 percent export directly and 7 percent have foreign ownership above 10 percent.

Table A.2 reports descriptive differences across backup types. Solar-only firms face outage exposure similar to generator-only firms, with typical outages of 3.27 and 2.68 hours respectively, but report lower finance obstacles and higher rates of exporting and website use. Firms with no backup report the lowest outage exposure, while exposure is higher among all backup holders. The raw differences therefore offer little evidence that solar and generator adopters differ primarily in outage exposure; instead, they point toward differences in the capability and finance characteristics associated

with solar adoption.

4.6. Empirical Strategy

The baseline specification relates each backup outcome to the two outage-exposure measures within a fixed-effects framework, given in Equation 8. For firm i in country c and sector s, the outcome y_{ics} is regressed on outage duration and frequency, a vector of firm controls X, and country and sector fixed effects.

$$y_{ics} = \alpha + \beta_1 \text{Duration}_{ics} + \beta_2 \text{Frequency}_{ics} + X'_{ics}\delta + \mu_c + \lambda_s + \varepsilon_{ics} \quad (8)$$

The country effects capture country-level differences across economies, including grid quality, fuel prices, solar irradiance, and renewable-energy policy. The exposure coefficients are identified from variation across firms within the same country and sector. The pooled specifications are estimated with survey weights rescaled such that each economy contributes equally. Binary outcomes are estimated as linear probability models and reproduced as logit average marginal effects in the appendix. The backup portfolio is estimated by multinomial logit with the same fixed effects and reported as average marginal effects. The sample contains 36 country clusters. Inference is based on standard errors clustered by country, with the wild-cluster bootstrap reported alongside as a robustness check that guards against the over-rejection conventional cluster-robust standard errors can produce when the number of clusters is small (Cameron et al., 2008; Roodman et al., 2019).

The estimates are read as conditional associations rather than causal effects, given that the data are a single cross-sectional wave that rules out firm fixed effects. Three sources of endogeneity could bias the exposure coefficients. The first is omitted variables. Local economic development, infrastructure quality, and firm characteristics that the survey does not record

could shape both a firm's outage exposure and its backup choice. Country and sector fixed effects absorb differences across economies and industries, the firm controls absorb observed differences across firms, and the instrumented specification is also reported with local-development controls. The second is measurement error. Outage frequency and duration are self-reported for a typical month, and classical error in these measures would attenuate the estimates toward zero. The third is reverse causality. A firm that has installed backup may notice or report outages differently, which links the outcome to the reported exposure. As a robustness check against all three sources, the paper instruments each firm's exposure with a leave-out mean, the average outage duration and frequency reported by all other firms in its country and locality. This isolates the grid-driven component of exposure, shared across nearby firms, from the firm's own circumstances and reporting. The exclusion restriction is that other firms' outages affect a firm's backup choice only through its own exposure, which fails if the shared local grid also proxies for local development. The specification is therefore reported with and without local-development controls, and the instrument serves as a directional check rather than a preferred estimate.

5. Results

5.1. The Asymmetric Response

The response of backup adoption to outage exposure is asymmetric across the two technologies. Table 2 reports the baseline specification with the generator and solar indicators as outcomes. The results show that an additional hour of typical outage duration is associated with a 0.011 increase in the probability of generator ownership, significant at the one percent level.

The corresponding estimate for solar adoption is far smaller, at 0.004, and only marginally significant. The asymmetry is therefore a difference in the strength of the two responses. Table A.3 confirms that the generator and solar duration slopes differ. The duration margin dominates the frequency margin, with the frequency association small and not significant. Among the controls, larger, older, and foreign-owned firms are more likely to run a generator, while larger and exporting firms are more likely to hold solar. Figure 3 shows the same asymmetry at the country level, with generator adoption rising with outage duration across economies and solar adoption rising at roughly half that rate.

5.2. Adoption Margins and Channels

The asymmetry holds for the adoption decision, for the share of electricity each technology supplies, and for broader energy practices. Table 3 extends the baseline across four outcomes. For the electricity shares, the estimates indicate that an additional hour of outage duration is associated with a 0.478 percentage-point increase in the share of electricity drawn from the generator, significant at the one percent level. In contrast, solar share shows no association with duration and a negative frequency association, significant at the five percent level. Active energy management shows a marginal duration association, significant only at the ten percent level, while carbon monitoring shows no association once sector and country fixed effects are included. The response to unreliability is therefore concentrated in dispatchable fossil capacity rather than spread across energy practices. A consistent secondary pattern reinforces the dispatchability findings. Outage frequency carries a small negative association with solar across the specifications—on the solar electricity share (-0.063), on developing-country solar adoption (-0.0009),

and more sharply in Sub-Saharan Africa (-0.0014).

5.3. The Backup Portfolio

Table 4 reports average marginal effects from a multinomial logit across four backup categories, namely no backup, generator only, solar only, and both. The findings suggest that outage duration is associated with a shift away from no backup and toward generator ownership. An additional hour of outage duration lowers the probability of holding no backup by 0.015 and raises the probability of holding a generator only by 0.011, both significant at the one percent level. The effect on solar-only adoption is small (0.001) and statistically insignificant, while the probability of holding both technologies rises by 0.002, significant at the one percent level. Figure 1 shows a similar pattern, with backup adoption increasing with outage exposure, predominantly through generator ownership.

5.4. Credit and the Generator Margin

Table 5 interacts outage exposure with the firm's own financing constraint and, separately, with the depth of its country's credit market. The evidence shows that most interaction terms are statistically insignificant. Notably, the significant interactions are concentrated on outage frequency. In the generator equation, the response to frequency is stronger in countries with deeper credit markets, with an interaction coefficient of 0.010, significant at the five percent level. The difference between the generator and solar frequency responses is also larger, at 0.009, while the firm's own financing constraint produces no comparable gap. The only significant own-constraint interaction is a small negative frequency effect on solar adoption (-0.002), with no corresponding effect on generator ownership. The generator-solar duration gap is

also larger in deeper credit markets, at 0.008, although the underlying duration interactions are not statistically significant. These results are consistent with the fixed-cost nature of generator investment. Deeper credit markets may enable firms facing frequent interruptions to respond through generator ownership, whereas the duration margin shows no comparable credit interaction. Among financially constrained firms, the weak solar response does not strengthen, and generator adoption is not more responsive to outage duration. The results therefore provide no evidence that greater access to credit shifts outage-exposed firms toward solar.

5.5. Capability, Capital, and Solar Adoption

Table 6 relates solar adoption to firm capability measures while controlling for outage exposure, allowing the two relationships to be examined jointly across three samples—the high-income benchmark, the full developing-country sample, and Sub-Saharan Africa. In the developing-country sample, the findings show that outage exposure is not statistically associated with solar adoption, whereas adoption increases with firm size, exporting, and website use, with marginal effects of 0.032, 0.063, and 0.043, respectively, all significant at the one percent level. Figure 4 illustrates the size asymmetry, with solar adopters having a substantially higher employment distribution than non-adopters. Figure 5 shows solar adoption by firm size, comparing developing economies with the high-income comparison group. Solar adoption rises from 10.3 percent among small firms to 14.8 percent among large firms in developing economies and from 8.2 to 25.6 percent in the high-income comparison group. Access to formal credit, measured by whether the firm holds a bank loan or line of credit, is not individually significant in any sample. The finance relationship therefore appears to

operate through broader firm capability and financial constraints rather than credit access. Table A.2 provides complementary descriptive evidence. Firms adopting both technologies are the largest and most capable, while solar-only firms, despite being smaller than generator-only firms, report the lowest finance obstacle and credit constraint and a higher export rate. Their outage exposure is nevertheless similar to that of generator-only firms.

Figure 2 provides corresponding cross-country evidence. Solar adoption shows no systematic relationship with outage incidence. Adoption rates are 27.6–35.5 percent in the Netherlands, Austria, and Germany, despite low outage incidence, compared with 0.6–1.4 percent in Ethiopia, Guinea, and Mozambique, where 58–91 percent of firms report outages (Table A.1). This pattern is inconsistent with outage exposure being the primary driver of solar adoption. The high-income benchmark provides a useful comparison as outages are rare and short, with typical firms reporting 0.0–0.3 outages per month, leaving limited scope for a necessity channel. In these economies, firm size and managerial experience are the capability measures significantly associated with solar adoption, while exporting and certification are positively associated but not significant. Managerial experience is negative and insignificant in developing economies but positive and significant in the high-income sample, while foreign ownership is negative in the latter and close to zero in the former. The evidence therefore does not imply a uniform set of determinants across settings. Nevertheless, firm capability, most consistently firm size, is positively associated with solar adoption in both high-outage and reliable-grid environments.

Table 7 examines the capital mechanism more directly through firms' investment and financing decisions. The first two columns show that a firm that purchased fixed assets in the previous year is 0.079 more likely to own a

generator, significant at the one percent level, but the association with solar adoption is not significant. Given that a generator is itself a fixed asset, the generator association is partly mechanical; the absence of a corresponding relationship for solar, which is also a fixed asset, is the informative part of the comparison. Firms undertaking lumpy capital expenditure appear to direct it toward dispatchable generation rather than intermittent solar, consistent with the dispatchability mechanism. The final two columns examine the financing of investment among firms that invested. Reliance on external investment finance rather than internal funds is associated with 0.028 higher solar adoption in developing economies, with no corresponding association in the high-income benchmark. The developing-country estimate is only marginally significant. The comparison across the two samples is therefore more informative than either coefficient, suggesting that external finance is more relevant to solar adoption where capital is relatively constrained.

5.6. Robustness

The asymmetry is robust across the specifications reported in the appendix. Table A.3 reports the difference between the generator and solar duration coefficients from the baseline specification without interactions. The difference is 0.007 and rejects equality of the generator and solar duration slopes at the five percent level. The 95 percent confidence interval for the solar duration coefficient, from -0.0009 to 0.0095, remains below the generator point estimate of 0.011. Table A.4 shows that the pattern holds across firm-size classes, while Table A.5 shows similar results across regions. The interaction terms are jointly insignificant in both specifications. Some individual interactions are significant with clustered standard errors, but none is jointly significant. A direct test of slope equality, reported as the

difference between the generator and solar duration coefficients in the Gap columns, rejects equality at the five percent level in the regional specification but not in the firm-size specification. Table A.6 considers alternative measures of outage exposure. Outage duration, the frequency–duration interaction, and an indicator for experiencing any outage are each positively and significantly associated with generator ownership, with no significant association with solar adoption. Outage frequency is positive but not significant. Firms experiencing any outage are 0.139 more likely to own a generator, with no corresponding increase in solar adoption. Table A.7 reproduces the binary outcomes using logit average marginal effects and yields the same asymmetry. Table A.8 excludes influential countries sequentially, including the country with the highest solar adoption and the three largest samples. The generator–duration association remains between 0.011 and 0.012 per additional hour, while the solar association remains small and, at most, marginally significant.

As a robustness check, Table A.9 examines omitted local conditions, measurement error in self-reported outages, and reverse causality using the average exposure reported by other firms in the same country and locality as instruments. The frequency instrument is strong, with a first-stage F-statistic of 126.9, whereas the duration instrument is weak, with a first-stage F-statistic of 4.9 and a joint Kleibergen–Paap rk F-statistic of 3.4. The IV estimates are therefore interpreted as a directional check rather than as preferred magnitudes. The instrumented duration coefficient for generator ownership remains positive and significant at the five percent level after controlling for local development, while the solar coefficient remains insignificant. The Durbin–Wu–Hausman test does not reject exogeneity for either outcome, providing no evidence of systematic endogeneity in the reported

outage measures. Overall, the appendix results consistently show a stronger relationship between outage exposure and generator ownership than between outage exposure and solar adoption.

6. Conclusions

This paper uses the 2025 and 2026 rounds of the World Bank Enterprise Surveys to study how firms in 36 developing economies choose between the diesel generator and solar self-generation when the grid is unreliable, with 17 high-income economies held out as a benchmark. The analysis separates two mechanisms. The first is dispatchability. A generator can be switched on at the moment of an outage and solar without storage is intermittent, which implies that outage exposure should track generator adoption and not solar adoption. The evidence supports this hypothesis. Outage exposure is associated with a higher probability of generator ownership, more strongly with the duration of outages than with their frequency. The association with solar adoption is weak, marginally significant at the ten percent level. Firms exposed to worse outages are more likely to hold both a generator and solar than to hold only solar. The second mechanism is capital. Solar requires a large outlay before it produces any power, which implies that its adoption should track capability and finance. The evidence in the analysis supports this. Solar adopters are larger, more likely to export, and less financially constrained, with firm size, the clearest capability measure, associated with solar adoption in the high-income benchmark, where outages are rare. A third result follows from combining the two mechanisms. Deeper credit does not move outage-exposed firms toward solar, consistent with the binding constraint being the capital needed to make solar dispatchable rather than

the price of panels.

The two results have different interpretations. The outage-response asymmetry is consistent with the importance of dispatchability but does not identify relative costs. Solar need not respond to outages even if its lifetime cost is low. The relationship between solar adoption and firm capability instead points to the role of upfront capital in determining adoption. Overall, the results describe two different forms of self-generation. Generators provide an immediate response to power interruptions, with costs incurred largely through fuel use. Solar represents a longer-term investment that requires sufficient capital to finance the initial outlay.

Several limitations qualify these conclusions. The data are cross-sectional and do not permit firm fixed effects or analysis of adoption over time. The estimates therefore represent conditional associations. The leave-out-mean instrument serves as a robustness check on omitted local conditions, measurement error, and reverse causality, but it does not establish causal effects. The solar measure records on-site solar self-generation rather than solar used specifically as backup, and the survey does not record battery storage. The role of dispatchability is therefore inferred from the characteristics of the technologies rather than directly observed. The sample also covers formal firms above a minimum size threshold and may under-represent the smallest and most financially constrained firms. The high-income sample is used as a benchmark and is not intended to represent developed economies. The generator and solar questions also differ in their framing, with the former referring to backup and the latter to annual self-generation. Future survey waves could provide a more direct test by recording battery capacity, grid connection, and the ability of solar systems to operate independently of the grid.

The policy implications concern the two constraints separately. Solar adoption may respond to financing arrangements that reduce or spread the upfront investment, including asset-based finance, leasing, pay-as-you-go arrangements, and partial credit guarantees. The results indicate that financing conditions remain relevant even as the cost of solar technology has fallen. At the same time, solar without storage cannot provide the same service as a generator during an outage. Large solar adoption therefore need not reduce generator use among firms with severe reliability problems. A cleaner substitute requires both solar generation and storage, with financing that addresses the upfront cost of the combined system. Table 7 supports this interpretation. External investment finance is positively associated with solar adoption among investing firms in developing economies, but not in the high-income benchmark. The results therefore point to financing for clean, dispatchable self-generation as a relevant margin for reducing firms' reliance on diesel while maintaining reliable electricity supply.

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Tables

Table 1: Summary Statistics

	Mean	SD	Min	Max
Panel A. Energy outcomes				
Has generator (=1)	0.400	0.490	0.00	1.00
Has solar (=1)	0.109	0.312	0.00	1.00
Generator % of electricity	10.722	21.417	0.00	100.00
Solar % of electricity	3.913	14.626	0.00	100.00
Energy management (=1)	0.318	0.466	0.00	1.00
CO_2 monitoring (=1)	0.134	0.340	0.00	1.00
Panel B. Outage exposure				
Outages per month	5.211	11.200	0.00	200.00
Outage duration (hrs)	2.031	3.575	0.00	72.00
Panel C. Firm characteristics				
Employees (ln)	2.584	1.023	0.00	9.80
Firm age (ln)	2.629	0.746	0.00	5.02
Exporter (=1)	0.085	0.279	0.00	1.00
Foreign-owned (>10%)	0.071	0.256	0.00	1.00
In capital city (=1)	0.506	0.500	0.00	1.00

Table 2: Outage Exposure and Backup Adoption

	Generator		Solar	
	(1)	(2)	(3)	(4)
Outage duration (hrs)	0.0111***	0.0111***	0.0042	0.0043*
	(0.0026)	(0.0026)	(0.0025)	(0.0025)
Outages per month	0.0011	0.0011	-0.0009	-0.0009
	(0.0022)	(0.0022)	(0.0006)	(0.0006)
Employees (ln)	0.1155***	0.1150***	0.0371***	0.0392***
	(0.0113)	(0.0114)	(0.0065)	(0.0065)
Firm age (ln)	0.0378***	0.0378***	-0.0069	-0.0067
	(0.0119)	(0.0118)	(0.0059)	(0.0059)
Exporter (=1)	-0.0349	-0.0362	0.0685***	0.0741***
	(0.0336)	(0.0337)	(0.0231)	(0.0237)
Foreign-owned (>10%)	0.0919***	0.0923***	0.0216	0.0196
	(0.0248)	(0.0250)	(0.0209)	(0.0212)
In capital city (=1)	0.0009	0.0014	0.0027	0.0000
	(0.0301)	(0.0301)	(0.0293)	(0.0291)
Constant	-0.0308	-0.0233	0.0191	-0.0174
	(0.0417)	(0.0520)	(0.0246)	(0.0335)
Observations	11,252	11,252	11,252	11,252
Country FE	Yes	Yes	Yes	Yes
Sector FE	No	Yes	No	Yes
Wild-cluster p (duration)		0.0005		0.0989
Wild-cluster p (frequency)		0.7367		0.2970
Adjusted R^2	0.2822	0.2822	0.1698	0.1719

Notes: Linear probability model; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table 3: Outage Exposure, Electricity Shares, and Green Practices

	Electricity share (%)		Green coping	
	Generator	Solar	Energy management	CO_2 monitoring
	(1)	(2)	(3)	(4)
Outage duration (hrs)	0.4776***	0.2462	0.0063*	-0.0017
	(0.1330)	(0.1893)	(0.0037)	(0.0026)
Outages per month	0.0158	-0.0625**	-0.0008	-0.0001
	(0.0995)	(0.0302)	(0.0013)	(0.0006)
Employees (ln)	2.8806***	0.5949**	0.0680***	0.0436***
	(0.5000)	(0.2668)	(0.0109)	(0.0103)
Firm age (ln)	0.0246	-0.3794	-0.0037	0.0027
	(0.5158)	(0.3386)	(0.0091)	(0.0070)
Exporter (=1)	-1.3142	3.7238***	0.0439	0.1171***
	(0.8696)	(1.3422)	(0.0423)	(0.0279)
Foreign-owned (>10%)	2.9244	0.0456	0.0733*	0.1059***
	(1.8258)	(0.6531)	(0.0373)	(0.0289)
In capital city (=1)	-1.1845	-0.0629	-0.0116	0.0151
	(1.5164)	(1.3110)	(0.0244)	(0.0133)
Constant	5.1714**	1.3595	0.1937***	0.0486
	(2.0974)	(1.6611)	(0.0455)	(0.0368)
Observations	11,252	11,252	11,252	11,252
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Wild-cluster p (duration)	0.0023	0.2911	0.1151	0.5465
Wild-cluster p (frequency)	0.9496	0.1483	0.7883	0.8901
Adjusted R^2	0.2856	0.1343	0.1324	0.1387

Notes: Cols (1)–(2) are OLS on the electricity share, the rest linear probability models; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table 4: Outage Exposure and the Backup Portfolio (Average marginal effect)

	Neither	Diesel only	Solar only	Both
	(1)	(2)	(3)	(4)
Outage duration (hrs)	-0.0146***	0.0111***	0.0013	0.0023***
	(0.0040)	(0.0031)	(0.0011)	(0.0007)
Outages per month	-0.0007	0.0011	-0.0002	-0.0003
	(0.0017)	(0.0013)	(0.0003)	(0.0006)
Employees (ln)	-0.1159***	0.0811***	0.0054	0.0293***
	(0.0112)	(0.0094)	(0.0044)	(0.0033)
Firm age (ln)	-0.0337***	0.0392***	-0.0038	-0.0017
	(0.0106)	(0.0132)	(0.0039)	(0.0054)
Exporter (=1)	0.0078	-0.0643**	0.0275***	0.0290***
	(0.0294)	(0.0314)	(0.0089)	(0.0099)
Foreign-owned (>10%)	-0.0940***	0.0713***	0.0075	0.0153
	(0.0219)	(0.0203)	(0.0128)	(0.0147)
In capital city (=1)	0.0019	-0.0019	-0.0012	0.0013
	(0.0379)	(0.0268)	(0.0127)	(0.0185)
Observations	11,252	11,252	11,252	11,252
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes

Notes: Multinomial-logit average marginal effects; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table 5: Outage Response by Firm and Country Credit Access

	Firm credit-constrained			Country credit-deep		
	Generator	Solar	Gap	Generator	Solar	Gap
	(1)	(2)	(3)	(4)	(5)	(6)
Outage duration (hrs)	0.0118***	0.0053*	0.0065	0.0103***	0.0022	0.0081**
	(0.0033)	(0.0031)	(0.0041)	(0.0029)	(0.0028)	(0.0031)
Outages per month	0.0018	-0.0001	0.0019	0.0003	-0.0009	0.0012
	(0.0025)	(0.0004)	(0.0023)	(0.0022)	(0.0007)	(0.0019)
Employees (ln)	0.1143***	0.0392***	0.0751***	0.1149***	0.0398***	0.0751***
	(0.0115)	(0.0066)	(0.0121)	(0.0114)	(0.0065)	(0.0120)
Firm age (ln)	0.0376***	-0.0065	0.0441***	0.0319***	-0.0068	0.0387***
	(0.0117)	(0.0059)	(0.0141)	(0.0106)	(0.0061)	(0.0135)
Exporter (=1)	-0.0364	0.0742***	-0.1106***	-0.0291	0.0701***	-0.0992**
	(0.0337)	(0.0235)	(0.0381)	(0.0339)	(0.0240)	(0.0377)
Foreign-owned (>10%)	0.0895***	0.0176	0.0719**	0.0889***	0.0194	0.0696**
	(0.0247)	(0.0207)	(0.0304)	(0.0250)	(0.0211)	(0.0301)
In capital city (=1)	0.0005	-0.0003	0.0008	0.0031	0.0007	0.0024
	(0.0298)	(0.0290)	(0.0276)	(0.0300)	(0.0291)	(0.0277)
Credit constrained	-0.0002	0.0153	-0.0155			
	(0.0222)	(0.0155)	(0.0284)			
Outage duration (hrs) × Credit constrained	-0.0015	-0.0021	0.0005			
	(0.0037)	(0.0030)	(0.0047)			
Outages per month × Credit constrained	-0.0016	-0.0019***	0.0003			
	(0.0012)	(0.0006)	(0.0015)			
Outage duration (hrs) × Credit-deep country				0.0002	0.0047	-0.0045
				(0.0069)	(0.0046)	(0.0074)
Outages per month × Credit-deep country				0.0097**	0.0003	0.0094**
				(0.0039)	(0.0011)	(0.0041)

Continued on next page

Table 5: *continued*

	Firm credit-constrained			Country credit-deep		
	Generator	Solar	Gap	Generator	Solar	Gap
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-0.0198	-0.0236	0.0038	-0.0109	-0.0255	0.0146
	(0.0550)	(0.0364)	(0.0620)	(0.0507)	(0.0335)	(0.0600)
Observations	11,252	11,252	11,252	11,129	11,129	11,129
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.2825	0.1732	0.1991	0.2834	0.1575	0.1849
Wild-cluster p (duration int.)	0.6741	0.5056	0.9169	0.9771	0.3657	0.5584
Wild-cluster p (freq int.)	0.0956	0.1547	0.8282	0.0290	0.8102	0.0258
Wild-cluster p (gap slope, duration)			0.1119			0.0318
Wild-cluster p (gap slope, frequency)			0.5344			0.5855

Notes: Outage exposure interacted with the firm credit constraint (cols 1–3) and above-median country credit depth (cols 4–6); credit constrained = fully or partially credit constrained; credit-deep = domestic credit to the private sector (percent of GDP) above the developing-sample median; Gap = generator-solar, SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Firm Capability and Solar Adoption

	Solar adoption (=1)		
	Advanced	Developing	SSA
	(1)	(2)	(3)
Outage duration (hrs)		0.0041	0.0026
		(0.0025)	(0.0026)
Outages per month		-0.0009*	-0.0014***
		(0.0005)	(0.0004)
Employees (ln)	0.0442***	0.0319***	0.0248***
	(0.0113)	(0.0064)	(0.0069)
Exporter (=1)	0.0063	0.0629***	0.0931**
	(0.0219)	(0.0218)	(0.0406)
Quality certification (=1)	0.0338	0.0287	0.0004
	(0.0269)	(0.0244)	(0.0464)
Has website (=1)	-0.0042	0.0432***	0.0545**
	(0.0087)	(0.0137)	(0.0192)
Manager experience (ln)	0.0247**	-0.0096	-0.0060
	(0.0094)	(0.0081)	(0.0148)
Foreign-owned (>10%)	-0.0384*	0.0105	0.0007
	(0.0212)	(0.0209)	(0.0277)
Has bank loan/credit line (=1)	0.0193	0.0214	0.0290
	(0.0145)	(0.0151)	(0.0313)
Constant	-0.0220	-0.0193	0.0133
	(0.0496)	(0.0296)	(0.0443)
Observations	9,360	11,252	6,135
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Adjusted R^2	0.1794	0.1775	0.1433
Wild-cluster p (duration)		0.1052	0.3671
Wild-cluster p (frequency)		0.2578	0.0950

Notes: Solar adoption on firm-capability covariates; col 1 is the 17 high-income economies, cols 2–3 developing and Sub-Saharan Africa, SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Capital Deployment and the Backup-Technology Choice

	Capital deployment		External finance → solar	
	Generator	Solar	Developing	Advanced
	(1)	(2)	(3)	(4)
Outage duration (hrs)	0.0104***	0.0041	0.0016	
	(0.0026)	(0.0024)	(0.0027)	
Outages per month	0.0010	-0.0010	-0.0001	
	(0.0022)	(0.0006)	(0.0007)	
Employees (ln)	0.1101***	0.0376***	0.0397***	0.0559***
	(0.0113)	(0.0067)	(0.0080)	(0.0159)
Firm age (ln)	0.0377***	-0.0065	-0.0028	0.0338**
	(0.0115)	(0.0059)	(0.0107)	(0.0137)
Exporter (=1)	-0.0418	0.0733***	0.0792***	0.0095
	(0.0323)	(0.0230)	(0.0257)	(0.0326)
Foreign-owned (>10%)	0.0896***	0.0192	0.0323	-0.0768**
	(0.0243)	(0.0212)	(0.0312)	(0.0303)
In capital city (=1)	-0.0028	-0.0004	-0.0385	-0.0664***
	(0.0300)	(0.0284)	(0.0243)	(0.0220)
Invested in fixed assets (=1)	0.0785***	0.0203		
	(0.0164)	(0.0169)		
Investment externally financed (=1)			0.0275*	-0.0030
			(0.0146)	(0.0175)
Constant	-0.0387	-0.0210	-0.0203	0.0056
	(0.0508)	(0.0342)	(0.0435)	(0.0616)
Observations	11,237	11,237	3,936	5,062
Country FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Wild-cluster p (capital measure)	0.0001	0.2621	0.0759	0.8664
Adjusted R^2	0.2870	0.1730	0.2255	0.1812

Notes: Cols 1–2 the full developing sample, cols 3–4 investors only; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Appendix A. Additional tables

Table A.1: Per-Country Adoption and Outage Exposure

Country	N	Solar %	Solar % of electricity	Generator %	Generator % of electricity	Outage incidence %	Outages per month	Loss % sales
Sub-Saharan Africa								
Burundi	151	7.7	14.8	31.8	37.5	93.4	5.5	1.9
Comoros	130	16.6	55.2	55.5	48.4	91.2	42.3	7.2
Ethiopia	867	1.4	14.9	55.3	14.4	86.7	6.6	3.3
Gabon	68	3.8	18.5	60.2	27.7	76.8	6.8	3.6
Guinea	159	0.6	19.0	50.5	18.1	91.0	3.1	0.1
Guinea Bissau	133	10.0	43.8	44.3	34.3	70.9	2.6	0.6
Kenya	992	9.4	21.2	60.4	14.5	74.8	2.8	2.7
Liberia	145	2.1	15.6	89.6	52.2	68.5	6.2	0.8
Malawi	132	11.2	12.8	35.3	13.7	88.3	8.7	1.3
Mauritania	146	10.3	23.8	46.0	19.6	82.5	4.3	1.0
Mozambique	582	1.0	35.3	20.5	17.0	58.0	2.7	2.2
Niger	127	8.6	36.9	72.4	21.8	76.3	7.0	1.9
Nigeria	872	22.7	39.7	84.9	51.3	81.3	10.8	5.4
Sao Tome And Principe	108	7.9	37.2	49.4	24.4	71.1	6.3	2.3
Somalia	217	8.9	26.3	12.3	12.3	48.5	1.5	1.3
Uganda	521	11.7	24.7	38.6	11.1	79.1	3.8	3.7
Zambia	463	31.3	37.6	55.8	48.7	88.6	21.3	5.1
Zimbabwe	322	45.5	39.0	53.9	26.8	53.6	3.9	1.9
East Asia & Pacific								
Australia	155	28.6	27.5	22.9	2.5	18.2	0.3	0.0
Brunei Darussalam	73	6.6	8.5	11.3	16.0	9.5	0.1	0.0
Fiji	92	6.5	11.6	36.2	26.4	82.7	2.0	1.2
Japan	1512	6.0	10.0	10.3	1.7	1.0	0.0	0.0
Kiribati	123	42.6	30.4	20.3	20.6	89.1	3.6	1.1
Mongolia	498	4.3	42.3	16.8	21.3	15.5	0.3	0.4
Solomon Islands	82	4.5	15.9	69.1	21.6	78.5	3.0	3.7
Thailand	565	4.3	47.1	0.9	62.3	6.8	0.1	0.1
South Asia								
Afghanistan	391	49.6	47.8	65.5	36.6	41.3	13.2	1.6
Maldives	93	5.9	28.7	13.4	49.7	58.8	1.1	1.3
Sri Lanka	436	8.8	62.3	17.9	31.7	29.7	0.9	0.2

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Table A.1: *continued*

Country	N	Solar %	Solar % of electricity	Generator %	Generator % of electricity	Outage incidence %	Outages per month	Loss % sales
Middle East & North Africa								
Djibouti	140	3.5	22.0	64.8	18.1	55.5	1.9	0.1
Egypt	938	0.0	16.1	5.2	12.0	35.2	4.2	1.1
Kuwait	140	0.0	.	0.2	35.9	0.0	0.0	0.0
Oman	171	0.3	16.3	7.4	45.8	18.4	0.3	0.1
Qatar	475	0.0	.	11.4	17.3	0.0	0.0	0.0
Saudi Arabia	980	0.1	22.9	1.9	31.5	0.0	0.0	0.0
Latin America & Caribbean								
Argentina	412	4.8	15.0	15.7	3.7	25.5	0.8	0.1
Belize	122	9.3	36.8	51.1	17.3	76.9	1.9	1.5
Bolivia	431	1.8	13.2	4.4	32.8	19.0	0.4	0.5
Chile	694	8.3	31.8	34.8	13.7	36.3	0.8	0.2
Guatemala	119	10.8	39.1	26.5	12.2	67.0	2.1	0.3
St Lucia	96	10.1	23.3	31.3	4.1	27.3	0.5	0.2
Suriname	131	4.2	25.3	23.7	21.3	57.8	1.3	1.9
Europe & Central Asia								
Albania	298	8.6	65.9	19.0	4.1	23.9	1.0	0.3
Austria	550	33.1	35.5	4.9	13.9	5.8	0.2	0.0
Denmark	489	10.2	29.2	12.3	8.4	3.5	0.0	0.0
Finland	348	7.4	17.3	7.8	1.4	7.4	0.1	0.0
France	726	6.2	26.9	10.4	2.5	13.0	0.3	0.1
Germany	1715	27.6	35.0	4.8	8.8	3.5	0.1	0.0
Kosovo	150	3.2	39.1	40.7	10.1	43.9	2.9	0.3
Lithuania	195	3.9	60.8	6.8	3.0	8.2	0.1	0.0
Netherlands	524	35.5	44.7	9.8	8.3	8.5	0.2	0.1
Norway	234	2.3	10.9	9.6	5.0	7.3	0.1	0.0
Switzerland	379	17.5	35.4	6.3	3.9	2.1	0.0	0.0
All								
All (pooled)	20612	11.1	34.9	30.4	25.0	44.5	3.6	1.0

Notes: Solar % of electricity and Generator % of electricity give the electricity share from each technology among the firms that use it, and are therefore conditional on adoption.

Table A.2: Firm Characteristics by Backup-Power Type

	Neither	Generator only	Solar only	Both
Outage duration (hrs)	1.39	2.68	3.27	3.19
Outages per month	3.57	7.06	7.41	7.79
Employees (ln)	2.36	2.85	2.52	3.27
Exporter (=1)	0.08	0.08	0.12	0.17
Finance obstacle (0–4)	1.70	1.66	1.43	1.57
Credit constrained (=1)	0.42	0.39	0.36	0.41
Has bank loan/credit line (=1)	0.28	0.33	0.28	0.37
Has website (=1)	0.51	0.56	0.57	0.66

Notes: Estimation-sample firms split by which backup technology the firm uses. Finance obstacle is coded 0 (no obstacle) to 4 (very severe obstacle). Credit constrained = fully or partially credit constrained.

Table A.3: The Generator-Solar Slope Difference and the Bound on the Solar Response

	Generator	Solar	Gap
	(1)	(2)	(3)
Outage duration (hrs)	0.0111***	0.0043*	0.0068**
	(0.0026)	(0.0025)	(0.0028)
Outages per month	0.0011	-0.0009	0.0020
	(0.0022)	(0.0006)	(0.0020)
Employees (ln)	0.1150***	0.0392***	0.0757***
	(0.0114)	(0.0065)	(0.0120)
Firm age (ln)	0.0378***	-0.0067	0.0445***
	(0.0118)	(0.0059)	(0.0142)
Exporter (=1)	-0.0362	0.0741***	-0.1103***
	(0.0337)	(0.0237)	(0.0380)
Foreign-owned (>10%)	0.0923***	0.0196	0.0727**
	(0.0250)	(0.0212)	(0.0300)
In capital city (=1)	0.0014	0.0000	0.0014
	(0.0301)	(0.0291)	(0.0278)
Constant	-0.0233	-0.0174	-0.0059
	(0.0520)	(0.0335)	(0.0608)
Observations	11,252	11,252	11,252
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Adjusted R^2	0.2822	0.1719	0.1992
Solar duration 95pct CI lower		-0.0009	
Solar duration 95pct CI upper		0.0095	
Wild-cluster p (gap slope, duration)			0.0283
Wild-cluster p (gap slope, frequency)			0.3975

Notes: Baseline (Table 2) spec for the generator, solar, and the generator-solar gap; the Gap column tests slope equality and the Solar column reports the 95% CI on the duration coefficient, SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.4: Firm-Size Heterogeneity

	Generator	Solar	Gap
	(1)	(2)	(3)
Outage duration (hrs)	0.0097***	0.0057*	0.0040
	(0.0026)	(0.0031)	(0.0030)
Outages per month	0.0007	-0.0007	0.0015
	(0.0024)	(0.0007)	(0.0020)
Employees (ln)	0.1171***	0.0394***	0.0777***
	(0.0130)	(0.0063)	(0.0141)
Firm age (ln)	0.0374***	-0.0067	0.0441***
	(0.0119)	(0.0060)	(0.0145)
Exporter (=1)	-0.0333	0.0725***	-0.1058***
	(0.0338)	(0.0233)	(0.0378)
Foreign-owned (>10%)	0.0932***	0.0197	0.0734**
	(0.0257)	(0.0215)	(0.0310)
In capital city (=1)	0.0029	-0.0001	0.0031
	(0.0305)	(0.0291)	(0.0279)
Medium firm (20–99)	0.0031	0.0101	-0.0070
	(0.0238)	(0.0135)	(0.0268)
Large firm (≥100)	-0.0636	0.0274	-0.0910*
	(0.0423)	(0.0310)	(0.0501)
Outage duration (hrs) × Medium firm (20–99)	0.0025	-0.0053	0.0077
	(0.0049)	(0.0034)	(0.0046)
Outages per month × Medium firm (20–99)	0.0012	-0.0006	0.0018
	(0.0025)	(0.0008)	(0.0022)
Outage duration (hrs) × Large firm (≥100)	0.0020	-0.0067	0.0087
	(0.0159)	(0.0063)	(0.0193)
Outages per month × Large firm (≥100)	0.0056	-0.0030	0.0086*

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Table A.4: *continued*

	Generator	Solar	Gap
	(1)	(2)	(3)
	(0.0038)	(0.0024)	(0.0049)
Constant	-0.0261	-0.0192	-0.0069
	(0.0576)	(0.0341)	(0.0667)
Observations	11,211	11,211	11,211
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Adjusted R^2	0.2853	0.1714	0.2009
Wild-cluster p (interactions, joint)	0.7240	0.1899	0.3153
Wild-cluster p (gap slope, duration)			0.2010
Wild-cluster p (gap slope, frequency)			0.5503

Notes: Outage exposure interacted with firm size (small is the reference); Gap = generator-solar, SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.5: Regional Heterogeneity

	Generator	Solar	Gap
	(1)	(2)	(3)
Outage duration (hrs)	0.0104***	0.0029	0.0074**
	(0.0028)	(0.0027)	(0.0030)
Outages per month	-0.0005	-0.0015***	0.0010
	(0.0022)	(0.0005)	(0.0023)
Employees (ln)	0.1146***	0.0390***	0.0756***
	(0.0113)	(0.0067)	(0.0120)
Firm age (ln)	0.0375***	-0.0067	0.0443***
	(0.0122)	(0.0057)	(0.0144)
Exporter (=1)	-0.0344	0.0737***	-0.1081***
	(0.0330)	(0.0237)	(0.0375)
Foreign-owned (>10%)	0.0931***	0.0206	0.0724**
	(0.0257)	(0.0211)	(0.0301)
In capital city (=1)	0.0050	0.0013	0.0037
	(0.0303)	(0.0294)	(0.0278)
Outage duration (hrs) × East Asia	0.0104	0.0164**	-0.0060
	(0.0075)	(0.0063)	(0.0119)
Outages per month × East Asia	0.0040	-0.0030	0.0070
	(0.0079)	(0.0035)	(0.0055)
Outage duration (hrs) × South Asia	-0.0215	-0.0091	-0.0124
	(0.0149)	(0.0056)	(0.0153)
Outages per month × South Asia	0.0043*	0.0024***	0.0019
	(0.0022)	(0.0005)	(0.0023)
Outage duration (hrs) × MENA	-0.0090**	-0.0061*	-0.0029
	(0.0038)	(0.0035)	(0.0049)

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Table A.5: *continued*

	Generator	Solar	Gap
	(1)	(2)	(3)
Outages per month × MENA	0.0078*	0.0015*	0.0063
	(0.0039)	(0.0009)	(0.0043)
Outage duration (hrs) × Latin America	-0.0004	-0.0016	0.0012
	(0.0082)	(0.0043)	(0.0090)
Outages per month × Latin America	0.0080	0.0045	0.0035
	(0.0072)	(0.0030)	(0.0049)
Outage duration (hrs) × Europe & Central Asia	0.0044	0.0209	-0.0165
	(0.0115)	(0.0133)	(0.0242)
Outages per month × Europe & Central Asia	0.0254*	-0.0035	0.0289
	(0.0143)	(0.0048)	(0.0190)
Constant	-0.0249	-0.0181	-0.0068
	(0.0510)	(0.0325)	(0.0604)
Observations	11,252	11,252	11,252
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes
Adjusted R^2	0.2862	0.1752	0.2010
Wild-cluster p (interactions, joint)	0.1742	0.7259	0.6080
Wild-cluster p (gap slope, duration)			0.0383
Wild-cluster p (gap slope, frequency)			0.6944
Wild-cluster p (duration x East Asia)		0.3008	

Notes: Outage exposure interacted with region (Sub-Saharan Africa is the reference); Gap = generator-solar, SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.6: The Generator-Solar Asymmetry across Alternative Outage Measures

	Generator (=1)				Solar (=1)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Outage duration (hrs)	0.0116***				0.0039			
	(0.0024)				(0.0025)			
Outages per month		0.0018				-0.0007		
		(0.0024)				(0.0007)		
Outages per month × Outage duration, hrs			0.0005**				0.0001	
			(0.0002)				(0.0001)	
Any outage (=1)				0.1388***				-0.0136
				(0.0312)				(0.0170)
Employees (ln)	0.1150***	0.1139***	0.1144***	0.1129***	0.0393***	0.0388***	0.0390***	0.0389***
	(0.0114)	(0.0115)	(0.0114)	(0.0114)	(0.0065)	(0.0065)	(0.0065)	(0.0065)
Firm age (ln)	0.0375***	0.0393***	0.0394***	0.0363***	-0.0064	-0.0061	-0.0058	-0.0057
	(0.0118)	(0.0118)	(0.0117)	(0.0112)	(0.0060)	(0.0059)	(0.0060)	(0.0060)
Exporter (=1)	-0.0369	-0.0383	-0.0376	-0.0357	0.0747***	0.0733***	0.0742***	0.0734***
	(0.0337)	(0.0346)	(0.0342)	(0.0324)	(0.0235)	(0.0240)	(0.0238)	(0.0238)
Foreign-owned (>10%)	0.0935***	0.0913***	0.0891***	0.0929***	0.0186	0.0192	0.0177	0.0185
	(0.0253)	(0.0246)	(0.0243)	(0.0223)	(0.0211)	(0.0210)	(0.0216)	(0.0208)
In capital city (=1)	0.0007	0.0005	-0.0003	0.0063	0.0006	-0.0003	0.0002	-0.0006
	(0.0299)	(0.0308)	(0.0302)	(0.0304)	(0.0289)	(0.0294)	(0.0290)	(0.0295)
Constant	-0.0232	-0.0106	-0.0144	-0.0797	0.0086	0.0211	0.0138	0.0246
	(0.0413)	(0.0433)	(0.0404)	(0.0474)	(0.0247)	(0.0239)	(0.0237)	(0.0267)
Observations	11,252	11,252	11,252	11,252	11,252	11,252	11,252	11,252
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.2819	0.2774	0.2788	0.2906	0.1714	0.1702	0.1701	0.1702

Notes: Each column regresses adoption on one outage measure; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.7: The Generator-Solar Asymmetry (Average Marginal Effect)

	Generator	Solar (=1)	Energy mgmt
	(1)	(2)	(3)
Outage duration (hrs)	0.0126***	0.0024*	0.0061
	(0.0033)	(0.0013)	(0.0037)
Outages per month	0.0008	-0.0005	-0.0010
	(0.0018)	(0.0004)	(0.0018)
Employees (ln)	0.1112***	0.0354***	0.0650***
	(0.0101)	(0.0056)	(0.0095)
Firm age (ln)	0.0378***	-0.0056	-0.0053
	(0.0123)	(0.0060)	(0.0092)
Exporter (=1)	-0.0304	0.0586***	0.0398
	(0.0311)	(0.0138)	(0.0387)
Foreign-owned (>10%)	0.0872***	0.0177	0.0681**
	(0.0223)	(0.0171)	(0.0347)
In capital city (=1)	-0.0005	0.0008	-0.0125
	(0.0291)	(0.0269)	(0.0254)
Observations	11,252	11,252	11,252
Country FE	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes

Notes: Average marginal effects from a firm-level logit; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.8: The Generator-Solar Asymmetry Excluding Influential Countries

	Generator (=1)			Solar (=1)		
	Full sample	Excluding Afghanistan	Excluding Egypt, Kenya & Nigeria	Full sample	Excluding Afghanistan	Excluding Egypt, Kenya & Nigeria
	(1)	(2)	(3)	(4)	(5)	(6)
Outage duration (hrs)	0.0111***	0.0118***	0.0120***	0.0043*	0.0048*	0.0055*
	(0.0026)	(0.0027)	(0.0034)	(0.0025)	(0.0026)	(0.0031)
Outages per month	0.0011	0.0005	0.0005	-0.0009	-0.0015***	-0.0009
	(0.0022)	(0.0025)	(0.0022)	(0.0006)	(0.0005)	(0.0006)
Employees (ln)	0.1150***	0.1160***	0.1191***	0.0392***	0.0392***	0.0382***
	(0.0114)	(0.0117)	(0.0118)	(0.0065)	(0.0067)	(0.0068)
Firm age (ln)	0.0378***	0.0389***	0.0434***	-0.0067	-0.0084	-0.0048
	(0.0118)	(0.0121)	(0.0126)	(0.0059)	(0.0058)	(0.0062)
Exporter (=1)	-0.0362	-0.0400	-0.0407	0.0741***	0.0706***	0.0772***
	(0.0337)	(0.0345)	(0.0369)	(0.0237)	(0.0240)	(0.0250)
Foreign-owned (>10%)	0.0923***	0.0922***	0.0983***	0.0196	0.0201	0.0168
	(0.0250)	(0.0256)	(0.0257)	(0.0212)	(0.0213)	(0.0223)
In capital city (=1)	0.0014	-0.0021	-0.0043	0.0000	-0.0030	0.0048
	(0.0301)	(0.0311)	(0.0329)	(0.0291)	(0.0301)	(0.0314)
Constant	-0.0233	-0.0332	-0.0515	-0.0174	-0.0111	-0.0272
	(0.0520)	(0.0539)	(0.0543)	(0.0335)	(0.0301)	(0.0370)
Observations	11,252	10,861	8,450	11,252	10,861	8,450
Adjusted R^2	0.2822	0.2837	0.2614	0.1719	0.1436	0.1768

Notes: The main result re-estimated dropping the highest-solar economy and then the three largest samples; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Table A.9: The Generator-Solar Asymmetry with Instrumented Outage Exposure

	Generator (=1)			Solar (=1)		
	OLS	IV	IV+loc	OLS	IV	IV+loc
	(1)	(2)	(3)	(4)	(5)	(6)
Outage duration (hrs)	0.0111***	0.0842**	0.0825**	0.0043	0.0629	0.0624
	(0.0026)	(0.0339)	(0.0349)	(0.0025)	(0.0539)	(0.0531)
Outages per month	0.0011	0.0029	0.0025	-0.0009	0.0056	0.0055
	(0.0022)	(0.0047)	(0.0048)	(0.0006)	(0.0046)	(0.0045)
Employees (ln)	0.1149***	0.1224***	0.1221***	0.0393***	0.0457***	0.0455***
	(0.0114)	(0.0116)	(0.0116)	(0.0065)	(0.0099)	(0.0099)
Firm age (ln)	0.0380***	0.0299**	0.0293**	-0.0067	-0.0117	-0.0120
	(0.0118)	(0.0138)	(0.0137)	(0.0059)	(0.0107)	(0.0106)
Exporter (=1)	-0.0368	-0.0188	-0.0219	0.0746***	0.0928***	0.0911***
	(0.0340)	(0.0337)	(0.0331)	(0.0238)	(0.0272)	(0.0271)
Foreign-owned (>10%)	0.0927***	0.0936***	0.0932***	0.0194	0.0148	0.0145
	(0.0250)	(0.0290)	(0.0289)	(0.0213)	(0.0304)	(0.0302)
In capital city (=1)	0.0011	0.0112	0.0143	0.0001	0.0115	0.0104
	(0.0301)	(0.0304)	(0.0316)	(0.0291)	(0.0322)	(0.0307)
Constant	-0.0237	0.0978	0.1652	-0.0176	0.2010	0.1867
	(0.0521)	(0.1019)	(0.1281)	(0.0335)	(0.1438)	(0.1991)
Observations	11,243	11,243	11,243	11,243	11,243	11,243
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Local-devt controls		No	Yes		No	Yes
First-stage F (duration)		4.9			4.9	
First-stage F (frequency)		126.9			126.9	
Kleibergen-Paap rk F		3.4			3.4	
Endogeneity (DWH) p		0.286			0.498	

Notes: Two-stage least squares instrumenting exposure with the leave-out mean exposure of other firms in the country×locality; SE clustered by country. * p<0.10, ** p<0.05, *** p<0.01.

Appendix B. Variable Definitions

Table B.1: Variable Definitions and Sources

Variable	Definition and coding	Source
Panel A. Energy outcomes		
Has generator (=1)	1 if the establishment owned or shared a generator in the last fiscal year, 0 otherwise	WBES c10
Has solar (=1)	1 if the establishment used on-site solar panels to generate electricity in the last fiscal year, 0 otherwise	WBES c43
Generator % of electricity	Percentage of the establishment's electricity from its own or shared generator, 0 to 100, with 0 for firms without a generator	WBES c11
Solar % of electricity	Percentage of the establishment's electricity from on-site solar, 0 to 100, with 0 for non-adopters	WBES c44
Backup portfolio	Four categories combining the two indicators, namely neither, generator only, solar only, or both	WBES c10, c43
Energy management (=1)	1 if the firm reports adopting energy management measures	WBES ge8d
CO_2 monitoring (=1)	1 if the firm monitors its own carbon dioxide emissions	WBES ge7
Panel B. Outage exposure		
Outages per month	Number of power outages in a typical month, with 0 for firms reporting no outages in the last fiscal year	WBES c6, c7
Outage duration (hrs)	Typical length of an outage in hours (hours plus minutes divided by 60), with 0 for firms reporting no outages	WBES c6, c8

Continued on next page

Table B.1: *continued*

Variable	Definition and coding	Source
Leave-out-mean instruments	Average outage duration and frequency reported by all other firms in the same country and locality	WBES c7, c8
Panel C. Firm characteristics		
Employees (ln)	Natural logarithm of permanent full-time employees at the end of the last fiscal year	WBES l1
Firm age (ln)	Natural logarithm of one plus the number of years the establishment has been operating	WBES b5
Exporter (=1)	1 if direct exports exceed 0 percent of the establishment's sales	WBES d3c
Foreign-owned (>10%)	1 if private foreign owners hold more than 10 percent of the firm	WBES b2b
In capital city (=1)	1 if the establishment is located in the capital city	WBES a3b
Quality certification (=1)	1 if the establishment holds an internationally recognised quality certification	WBES b8
Has website (=1)	1 if the establishment has its own website	WBES c22b
Manager experience (ln)	Natural logarithm of one plus the top manager's years of experience in the sector	WBES b7
Trains workers (=1)	1 if the establishment ran formal training programmes for permanent full-time employees	WBES l10
Panel D. Finance		
Credit constrained (=1)	1 if fully or partially credit constrained, 0 if unconstrained (definitions in Section 4.4)	ES Indicators
Has bank loan/credit line (=1)	1 if the establishment has a loan or line of credit from a financial institution	ES Indicators
Invested in fixed assets (=1)	1 if the establishment purchased fixed assets in the last fiscal year	WBES k4
Investment externally financed (=1)	Among firms that invested, 1 if less than 100 percent of fixed-asset purchases was financed internally	ES Indicators

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Table B.1: *continued*

Variable	Definition and coding	Source
Finance obstacle (0–4)	Degree to which access to finance is an obstacle, 0 no obstacle, 1 minor, 2 moderate, 3 major, 4 very severe	WBES k30
Country credit-deep (=1)	1 if domestic credit to the private sector (percent of GDP), most recent year, is above the developing-sample median	WDI

Notes: WBES codes refer to questions in the Enterprise Surveys standard questionnaire. ES Indicators = World Bank Enterprise Surveys Indicators Database. WDI = World Development Indicators, series FS.AST.PRVT.GD.ZS.

Figures

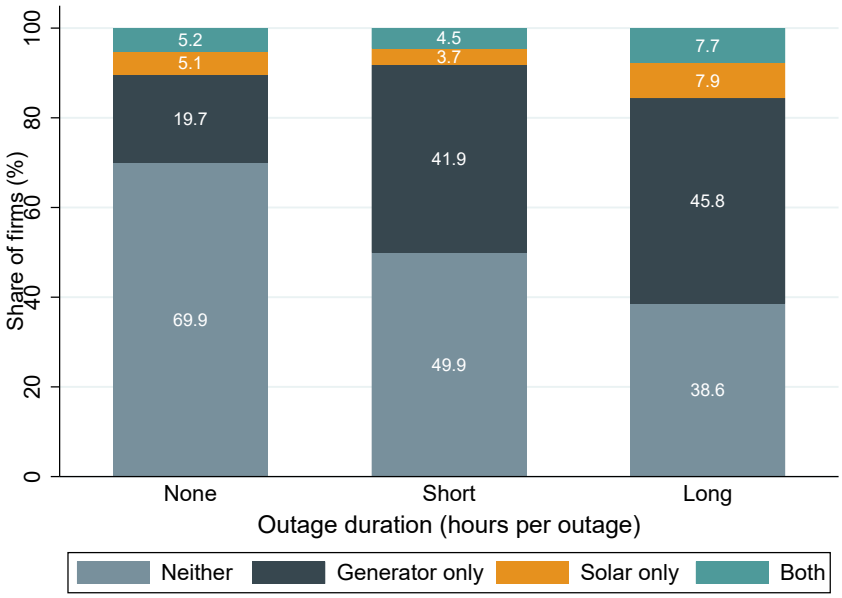


Figure 1. Backup-power adoption by level of outage exposure.

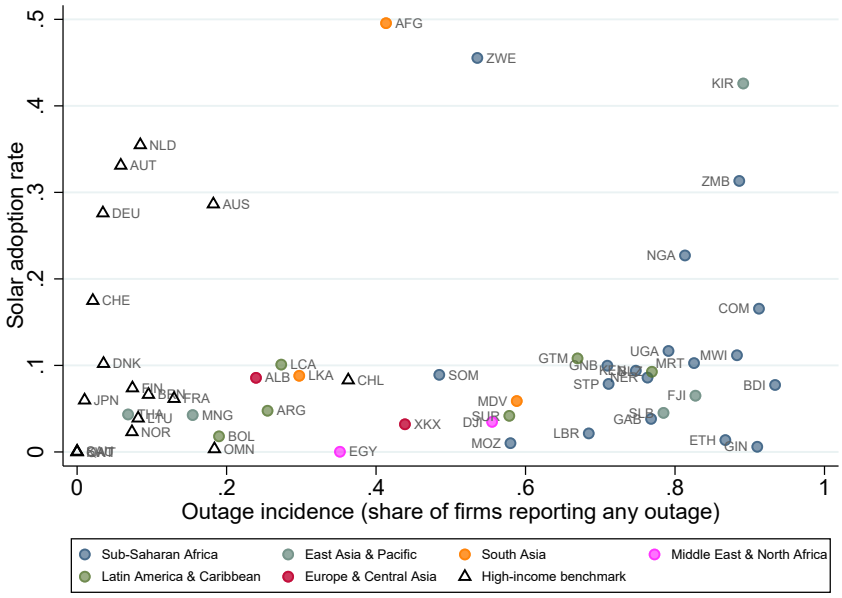


Figure 2. Solar adoption and grid reliability across economies.

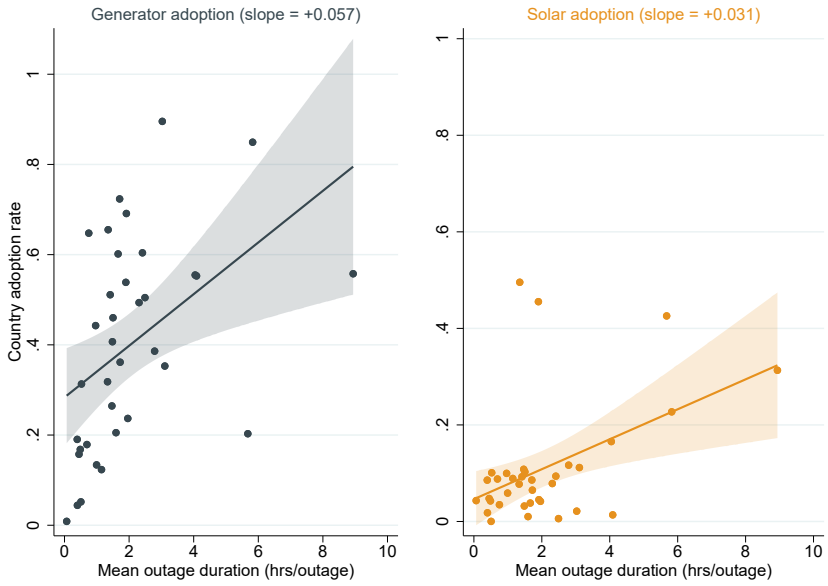


Figure 3. Generator and solar adoption against outage duration, by country.

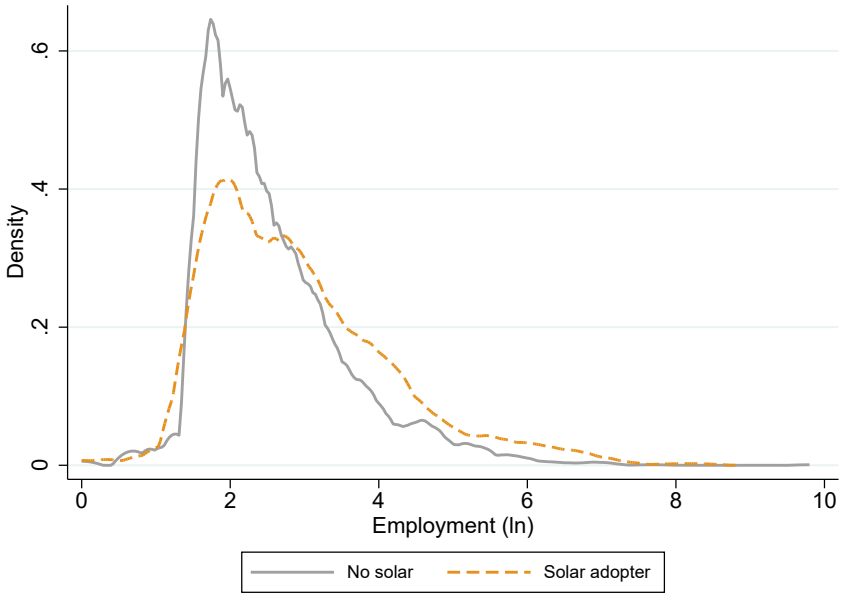


Figure 4. Distribution of firm size by solar adoption.

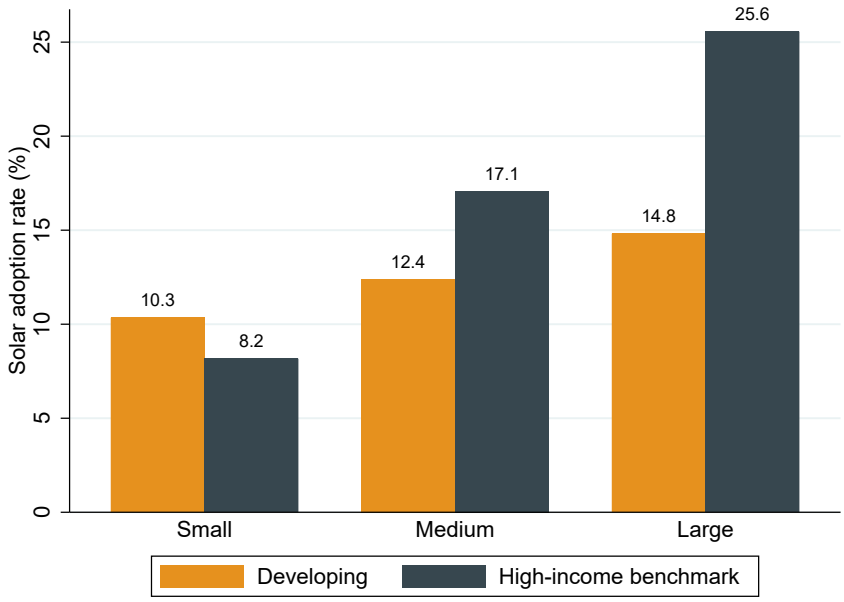


Figure 5. Solar adoption by firm-size class, developing and high-income benchmark.